

Problem 1. *Compute the following matrix arithmetic.*

1.

$$\begin{pmatrix} 4 & 7 & 8 \\ 8 & 10 & 0 \\ -5 & 13 & 22 \end{pmatrix} + \begin{pmatrix} 2 & 12 & 3 \\ 6 & 4 & 3 \\ 2 & -9 & 0 \end{pmatrix} = \begin{pmatrix} 6 & 19 & 11 \\ 14 & 14 & 3 \\ -3 & 4 & 22 \end{pmatrix}$$

2.

$$\begin{pmatrix} 2 & -2 & 2 \\ 6 & 8 & 5 \\ 0 & 23 & 22 \end{pmatrix} + \begin{pmatrix} 2 & 2 & -2 \\ -6 & -4 & -5 \\ 0 & -23 & -18 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = 4 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

3.

$$\begin{pmatrix} 5 & 44 \\ 0 & -2 \end{pmatrix} + \begin{pmatrix} 9 & -52 \\ -3 & -4 \end{pmatrix} = \begin{pmatrix} 14 & -8 \\ -3 & -6 \end{pmatrix}$$

4.

$$\begin{pmatrix} 22 & 1 \\ 7 & -6 \end{pmatrix} - \begin{pmatrix} -1 & 5 \\ 3 & -2 \end{pmatrix} = \begin{pmatrix} 23 & -4 \\ 4 & -4 \end{pmatrix}$$

5.

$$\begin{pmatrix} 21 & -5 & 3 \\ 6 & 10 & 6 \\ 0 & -7 & 12 \end{pmatrix} - \begin{pmatrix} 12 & -14 & -6 \\ -3 & 1 & -3 \\ -9 & -16 & 3 \end{pmatrix} = \begin{pmatrix} 9 & 9 & 9 \\ 9 & 9 & 9 \\ 9 & 9 & 9 \end{pmatrix} = 9 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Problem 2. *Matrix Multiplication*

1.

$$\begin{pmatrix} 4 & 7 & 8 \\ 8 & 10 & 0 \\ -5 & 13 & 22 \end{pmatrix} \times \begin{pmatrix} 2 & 1 & 3 \\ 6 & 4 & 3 \\ 2 & -4 & 0 \end{pmatrix} = \begin{pmatrix} 8+42+16 & 4+28-32 & 12+21+0 \\ 16+60+0 & 8+40+0 & 24+30+0 \\ -10+78+44 & -5+52-88 & -15+39+0 \end{pmatrix} \\ = \begin{pmatrix} 66 & 0 & 33 \\ 76 & 48 & 54 \\ 112 & -41 & 24 \end{pmatrix}$$

2.

$$\begin{pmatrix} 5 & 0 & 1 \\ 0 & 5 & 0 \\ 1 & 0 & 5 \end{pmatrix} \times \begin{pmatrix} 5 & 0 & 1 \\ 0 & 5 & 0 \\ 1 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 25+0+1 & 0+0+0 & 5+0+5 \\ 0+0+0 & 0+25+0 & 0+0+0 \\ 5+0+5 & 0+0+0 & 1+0+25 \end{pmatrix} \\ = \begin{pmatrix} 26 & 0 & 10 \\ 0 & 25 & 0 \\ 10 & 0 & 26 \end{pmatrix}$$

3.

$$\begin{pmatrix} 5 & 21 \\ 0 & -2 \end{pmatrix} \times \begin{pmatrix} 9 & -5 \\ -3 & -4 \end{pmatrix} = \begin{pmatrix} 45-63 & -25-84 \\ 0+6 & 0+8 \end{pmatrix} = \begin{pmatrix} -18 & -109 \\ 6 & 8 \end{pmatrix}$$

4.

$$\begin{aligned}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \times \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} &= \begin{pmatrix} -1-1 & 1+1 \\ 1+0 & -1-1 \end{pmatrix} \\ &= \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} = 2 \times \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\end{aligned}$$

Note: Eigen vector equation

5.

$$\begin{pmatrix} 4 & -5 & 3 \\ 6 & 10 & 2 \end{pmatrix} \times \begin{pmatrix} 9 & -1 \\ -3 & 1 \\ -9 & 0 \end{pmatrix} = \begin{pmatrix} 36+15-27 & -4-5 \\ 54-30-18 & -6+10+0 \end{pmatrix} = \begin{pmatrix} 24 & -9 \\ 6 & 4 \end{pmatrix}$$

Problem 3. Find the matrix X

1.

$$\begin{aligned}\frac{3}{2}X + \begin{pmatrix} -1 & 3 \\ 2 & -2 \end{pmatrix} &= \begin{pmatrix} 3 & -4 \\ 5 & 4 \end{pmatrix} \\ \frac{3}{2}X &= \begin{pmatrix} 3 & -4 \\ 5 & 4 \end{pmatrix} - \begin{pmatrix} -1 & 3 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 4 & -7 \\ 3 & 6 \end{pmatrix} \\ X &= \frac{2}{3} \begin{pmatrix} 4 & -7 \\ 3 & 6 \end{pmatrix}\end{aligned}$$

2.

$$\begin{aligned}X + \begin{pmatrix} 5 & 2 & -1 \\ -9 & 0 & -5 \\ 0 & 3 & -8 \end{pmatrix} &= \begin{pmatrix} 8 & 2 & -4 \\ 12 & 3 & 0 \\ -7 & 3 & -8 \end{pmatrix} \\ X &= \begin{pmatrix} 8 & 2 & -4 \\ 12 & 3 & 0 \\ -7 & 3 & -8 \end{pmatrix} - \begin{pmatrix} 5 & 2 & -1 \\ -9 & 0 & -5 \\ 0 & 3 & -8 \end{pmatrix} \\ X &= \begin{pmatrix} 3 & 0 & -3 \\ 21 & 3 & 5 \\ -7 & 0 & 0 \end{pmatrix}\end{aligned}$$

3.

$$\begin{aligned}X \times \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} &= \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = 2I \times \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \\ \text{therefore } X &= \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\end{aligned}$$

Note: We don't talk about matrix division, instead we attempt to find the inverse of a matrix

4.

$$\begin{aligned}
 A &= \begin{pmatrix} 5 & 4 \\ 7 & -5 \end{pmatrix} \\
 A^{-1} &= \frac{1}{\det[A]} \begin{pmatrix} -5 & -4 \\ -7 & 5 \end{pmatrix} \\
 \det[A] &= -25 - 28 = -53 \\
 A^{-1} &= \frac{1}{-53} \begin{pmatrix} -5 & -4 \\ -7 & 5 \end{pmatrix} \\
 A \cdot A^{-1} &= A^{-1} \cdot A = I
 \end{aligned}$$

Check that the matrix multiplied by its inverse is a Unitary matrix

5.

$$\begin{aligned}
 A &= \begin{pmatrix} 3/5 & -7/10 \\ -1/5 & 2/5 \end{pmatrix} \\
 A^{-1} &= \frac{1}{\det[A]} \begin{pmatrix} 2/5 & 7/10 \\ 1/5 & 3/5 \end{pmatrix} \\
 \det[A] &= 6/25 - 7/50 = 1/10 \\
 A^{-1} &= 10 \begin{pmatrix} 2/5 & 7/10 \\ 1/5 & 3/5 \end{pmatrix} = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix} \\
 A \cdot A^{-1} &= I
 \end{aligned}$$

Check that the matrix multiplied by its inverse is a Unitary matrix

Problem 4. *Some more matrices*

1.

$$\begin{aligned}
 A \cdot B &= \begin{pmatrix} 5 & 3 & 2 \\ 2 & 4 & 1 \\ 8 & 0 & 6 \end{pmatrix} \times \begin{pmatrix} 3 & -4 & -1 \\ 5 & 4 & 2 \\ -1 & 2 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 15 + 15 - 2 & -20 + 12 + 4 & -5 + 6 + 0 \\ 6 + 20 - 1 & -8 + 16 + 2 & -2 + 8 + 0 \\ 24 + 0 - 6 & -32 + 0 + 12 & -8 + 0 + 0 \end{pmatrix} \\
 &= \begin{pmatrix} 28 & -4 & 1 \\ 25 & 10 & 6 \\ 18 & -20 & -8 \end{pmatrix} \\
 B \cdot A &= \begin{pmatrix} 3 & -4 & -1 \\ 5 & 4 & 2 \\ -1 & 2 & 0 \end{pmatrix} \times \begin{pmatrix} 5 & 3 & 2 \\ 2 & 4 & 1 \\ 8 & 0 & 6 \end{pmatrix} \\
 &= \begin{pmatrix} 15 - 8 - 8 & 9 - 16 + 0 & 6 - 4 - 6 \\ 25 + 8 + 16 & 15 + 16 + 0 & 10 + 4 + 12 \\ -5 + 4 + 0 & -3 + 8 + 0 & -2 + 2 + 0 \end{pmatrix} \\
 &= \begin{pmatrix} -1 & -7 & -4 \\ 49 & 31 & 26 \\ -1 & 5 & 0 \end{pmatrix} \\
 A \cdot B &\neq B \cdot A
 \end{aligned}$$

2. Find the determinants of the above matrices $|A|$ and $|B|$
Show that

$$|AB| = |A||B|$$

$$|A| = 5(24 - 0) - 3(12 - 8) + 2(0 - 32) = 44$$

$$|B| = 3(0 - 4) - 4(0 + 2) - 1(10 + 4) = -18$$

$$|A||B| = 44 \times -18 = -792$$

$$|AB| = 28(-80 + 120) + 4(-200 - 108) + 1(-500 - 180) = -792$$

$$\text{Therefore } |AB| = |A||B|$$

3.

$$A(B + C) = A \left(\begin{pmatrix} 3 & -4 & -1 \\ 5 & 4 & 2 \\ -1 & 2 & 0 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 5 \\ 0 & 2 & 3 \\ 1 & 8 & 4 \end{pmatrix} \right)$$

$$= \begin{pmatrix} 5 & 3 & 2 \\ 2 & 4 & 1 \\ 8 & 0 & 6 \end{pmatrix} \times \begin{pmatrix} 5 & -4 & 4 \\ 5 & 6 & 5 \\ 0 & 10 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 40 & 18 & 43 \\ 30 & 26 & 32 \\ 40 & 28 & 56 \end{pmatrix}$$

$$AB + AC = \begin{pmatrix} 28 & -4 & 1 \\ 25 & 10 & 6 \\ 18 & -20 & -8 \end{pmatrix} + \begin{pmatrix} 12 & 22 & 42 \\ 5 & 16 & 26 \\ 22 & 48 & 64 \end{pmatrix}$$

$$= \begin{pmatrix} 40 & 18 & 43 \\ 30 & 26 & 32 \\ 40 & 28 & 56 \end{pmatrix}$$

$$A(B + C) = AB + AC$$

4.

$$(A + B) = \begin{pmatrix} 8 & -1 & 1 \\ 7 & 8 & 3 \\ 7 & 2 & 6 \end{pmatrix}$$

$$(A + B)^T = \begin{pmatrix} 8 & 7 & 7 \\ -1 & 8 & 2 \\ 1 & 3 & 6 \end{pmatrix}$$

$$A^T + B^T = \begin{pmatrix} 5 & 2 & 8 \\ 3 & 4 & 0 \\ 2 & 1 & 6 \end{pmatrix} + \begin{pmatrix} 3 & 5 & -1 \\ -4 & 4 & 2 \\ -1 & 2 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 7 & 7 \\ -1 & 8 & 2 \\ 1 & 3 & 6 \end{pmatrix}$$

$$(A + B)^T = A^T + B^T$$

Consider the transpose of the product of A and B

$$\begin{aligned}
 (AB)^T &= \begin{pmatrix} 28 & -4 & 1 \\ 25 & 10 & 6 \\ 18 & -20 & -8 \end{pmatrix}^T \\
 &= \begin{pmatrix} 28 & 25 & 18 \\ -4 & 10 & -20 \\ 1 & 6 & -8 \end{pmatrix} \\
 B^T A^T &= \begin{pmatrix} 3 & 5 & -1 \\ -4 & 4 & 2 \\ -1 & 2 & 0 \end{pmatrix} \times \begin{pmatrix} 5 & 2 & 8 \\ 3 & 4 & 0 \\ 2 & 1 & 6 \end{pmatrix} \\
 &= \begin{pmatrix} 15 + 15 - 2 & 6 + 20 + -1 & 24 + 0 + -6 \\ -20 + 12 + 4 & -8 + 16 + 2 & -32 + 0 + 12 \\ -5 + 6 + 0 & -2 + 8 + 0 & -8 + 0 + 0 \end{pmatrix} = \begin{pmatrix} 28 & 25 & 18 \\ -4 & 10 & -20 \\ 1 & 6 & -8 \end{pmatrix} \\
 (AB)^T &= B^T A^T
 \end{aligned}$$

5. The product of all 3 matrices

$$\begin{aligned}
 ABC &= (AB)C = A(BC) \\
 &= \begin{pmatrix} 57 & 0 & 132 \\ 56 & 68 & 179 \\ 28 & -104 & -2 \end{pmatrix}
 \end{aligned}$$

Problem 5. 1. Inverse of A

$$\begin{aligned}
 A &= \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix} \\
 A^{-1} &= \frac{1}{|A|} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} \\
 \det|A| &= 4 \times 6 - 7 \times 2 = 10 \\
 A^{-1} &= \frac{1}{10} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}
 \end{aligned}$$

2. Check $AA^{-1} = I$

$$\begin{aligned}
 AA^{-1} &= \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix} \frac{1}{10} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} \\
 &= \frac{1}{10} \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} \\
 &= \frac{1}{10} \begin{pmatrix} 24 - 14 & -28 + 28 \\ 12 - 12 & -14 + 24 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \\
 A^{-1}A &= \frac{1}{10} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix} \\
 &= \frac{1}{10} \begin{pmatrix} 24 - 14 & 42 - 42 \\ -8 + 8 & -14 + 24 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I
 \end{aligned}$$

Therefore:

$$AA^{-1} = A^{-1}A = I$$

3. Eigen Values & Eigen vectors

$$A = \begin{pmatrix} 11 & 10 \\ 1 & 2 \end{pmatrix}$$

$$A\vec{V} = \lambda\vec{V}$$

Characteristic Equation

$$|A - \lambda I| = 0$$

$$\left| \begin{pmatrix} 11 & 10 \\ 1 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} 11 - \lambda & 10 \\ 1 & 2 - \lambda \end{pmatrix} \right| = 0$$

$$(11 - \lambda)(2 - \lambda) - 10 = 0$$

$$\lambda^2 - 13\lambda + 12 = 0$$

$$(\lambda - 12)(\lambda - 1) = 0$$

$$\lambda = 1, 12$$

To find eigen vectors for each eigen value $\lambda = 1$

$$\begin{pmatrix} 11 & 10 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = 1 \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$\begin{pmatrix} 11X + 10Y \\ X + 2Y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X + Y = 0$$

$$X = -Y$$

eigen vector for $\lambda = 1$

$$\begin{pmatrix} X \\ -X \end{pmatrix}$$

To find eigen vectors for each eigen value $\lambda = 12$

$$\begin{pmatrix} 11 & 10 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = 12 \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$\begin{pmatrix} 11X + 10Y \\ X + 2Y \end{pmatrix} = \begin{pmatrix} 12X \\ 12Y \end{pmatrix}$$

$$X - 10Y = 0$$

$$X = 10Y$$

eigen vector for $\lambda = 12$

$$\begin{pmatrix} X \\ X/10 \end{pmatrix} \text{ Or } \begin{pmatrix} 10X \\ X \end{pmatrix}$$

4. Finding eigen values & eigen vectors of a 3×3 matrix

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 4 & 5 \\ 0 & 4 & 3 \end{pmatrix}$$

Characteristic Equation

$$\begin{aligned}
 |A - \lambda I| &= 0 \\
 \left| \begin{pmatrix} 2 & 0 & 0 \\ 0 & 4 & 5 \\ 0 & 4 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right| &= 0 \\
 \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & 4-\lambda & 5 \\ 0 & 4 & 3-\lambda \end{vmatrix} &= 0 \\
 2-\lambda \begin{vmatrix} 4-\lambda & 5 \\ 4 & 3-\lambda \end{vmatrix} - 0 \begin{vmatrix} 0 & 5 \\ 0 & 3-\lambda \end{vmatrix} + 0 \begin{vmatrix} 0 & 4-\lambda \\ 0 & 4 \end{vmatrix} &= 0 \\
 (2-\lambda)[(4-\lambda)(3-\lambda) - 20] &= 0 \\
 (2-\lambda)[\lambda^2 - 7\lambda - 8] &= 0 \\
 (2-\lambda)(\lambda - 8)(\lambda + 1) &= 0 \\
 \lambda &= -1, 2, 8
 \end{aligned}$$

Problem 6. 1. *Transpose of $R : R^T$*

$$R^T = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

2. *Inverse of $R : R^{-1}$*

$$R^{-1} = \frac{1}{\det|R|} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

$$\det|R| = \cos^2\theta + \sin^2\theta = 1$$

Therefore,

$$R^{-1} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Therefore, $R^T = R^{-1}$

Using the rotation matrix on $A = (2, 5)$

1. *Anti-clockwise rotation*

$$\begin{aligned}
 \begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} \cos(90^\circ) & -\sin(90^\circ) \\ \sin(90^\circ) & \cos(90^\circ) \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\
 \begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} -5 \\ 2 \end{pmatrix}
 \end{aligned}$$

2. *Clockwise rotation*

$$\begin{aligned}
 \begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} \cos(-90^\circ) & -\sin(-90^\circ) \\ \sin(-90^\circ) & \cos(-90^\circ) \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\
 \begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} 5 \\ -2 \end{pmatrix}
 \end{aligned}$$

3. *Anti-clockwise 180°*

$$\begin{aligned}\begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} \cos(180^\circ) & -\sin(180^\circ) \\ \sin(180^\circ) & \cos(180^\circ) \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\ \begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} -2 \\ -5 \end{pmatrix}\end{aligned}$$

and Clock-wise 180°

$$\begin{aligned}\begin{pmatrix} x'' \\ y'' \end{pmatrix} &= \begin{pmatrix} \cos(-180^\circ) & -\sin(-180^\circ) \\ \sin(-180^\circ) & \cos(-180^\circ) \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\ \begin{pmatrix} x'' \\ y'' \end{pmatrix} &= \begin{pmatrix} -2 \\ -5 \end{pmatrix} \\ (x', y') &\equiv (x'', y'')\end{aligned}$$